Titanic: Exploratory Data Analysis

Business Insights & Recommendations Report

Project 2 — Data Analytics

Dataset: Cleaned Titanic Passenger Records (891 passengers)

1. Data Understanding

This dataset contains passenger records from the RMS Titanic, the ship that sank in 1912 after colliding with an iceberg. The cleaned dataset contains **891 rows and 13 columns**, each row representing one passenger. The features include a mix of demographic information (age, sex), travel details (passenger class, fare, ticket number, cabin deck, port of embarkation), family relationship counts (siblings/spouses aboard, parents/children aboard), and the target variable (survived: Yes/No). The dataset has already been cleaned — there are no missing values and no duplicate rows — making it well-suited for direct exploratory analysis. Overall, this dataset provides enough information to explore how demographic and socio-economic factors (class, fare, gender, age, family size) related to a passenger's chances of survival.

Metric	Value
Total Passengers	891
Total Columns	13 (+4 engineered features)
Missing Values	0
Duplicate Rows	0
Overall Survival Rate	38.4%

2. Feature Engineering

Four new features were engineered from the existing columns to enrich the analysis:

Family Size (SibSp + Parch + 1)

Combines the separate family-count columns into one intuitive measure representing the total number of family members traveling together, including the passenger.

Is Alone (Yes / No flag based on Family Size)

Identifies whether a passenger was traveling alone or with family/group using a simple Yes/No flag, easier to visualize and compare than raw family-size values.

Age Group (Child / Teenager / Adult / Senior Citizen)

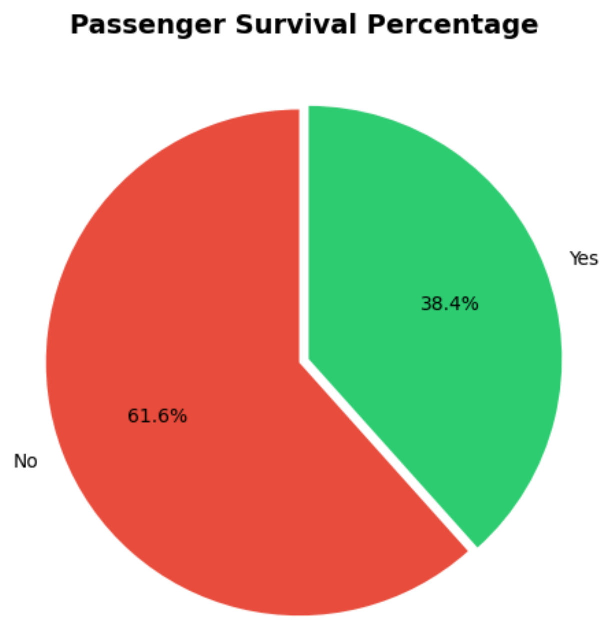
Divides passengers into life-stage categories instead of individual age values, making aggregate comparisons and survival patterns easier to identify.

Fare Category (Low / Medium / High Fare)

Groups passengers according to the ticket fare they paid instead of individual fare amounts, since fare is a continuous and highly skewed variable; also serves as a proxy for socio-economic status.

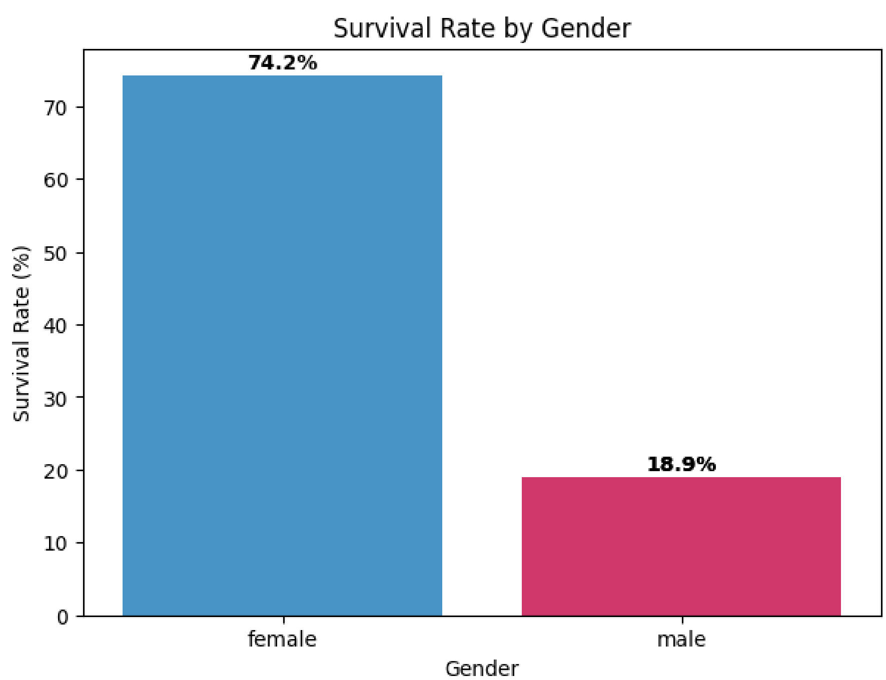
3. Exploratory Data Analysis

Q1: Overall Survival Rate



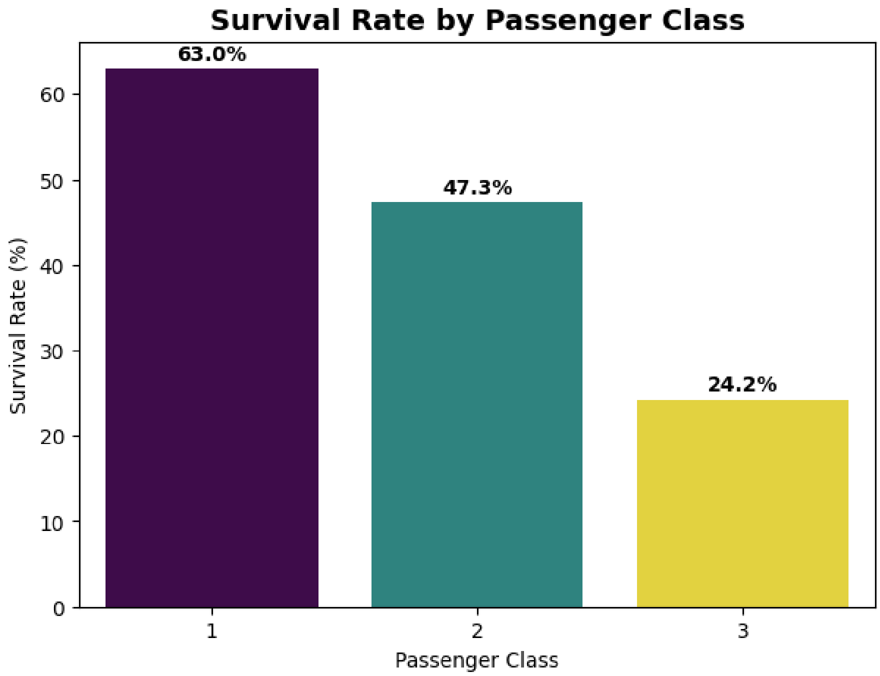
Only about 38.4% of passengers survived, while 61.6% did not survive. This shows that the Titanic disaster was catastrophic, with fewer than 2 in 5 passengers surviving overall.

Q2: Survival Rate by Gender



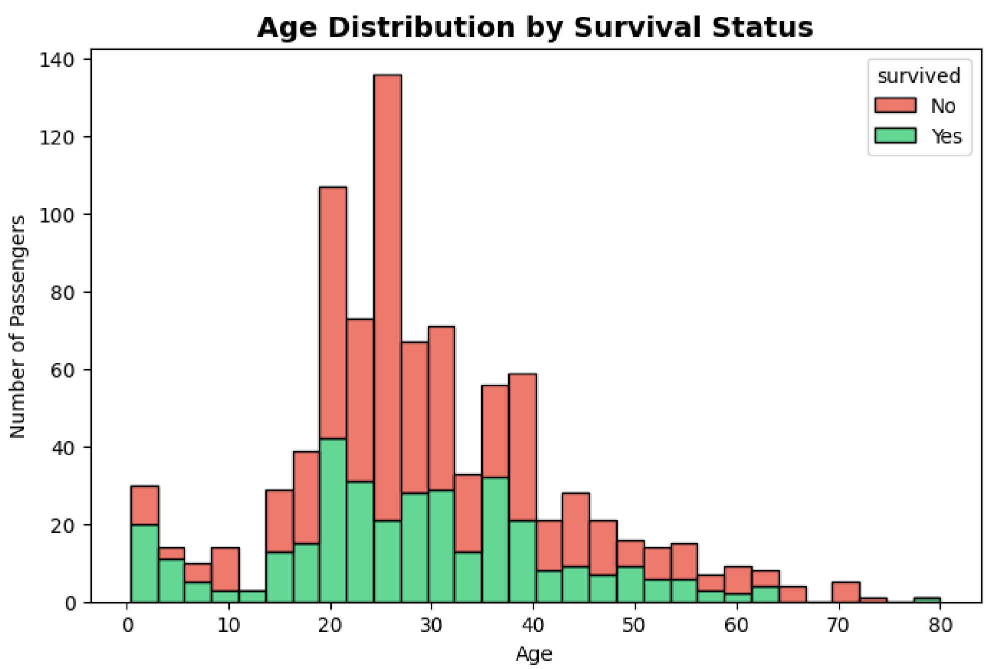
Female passengers had a much higher survival rate (74.2%) than male passengers (18.9%). This large difference suggests that women were given higher priority during the evacuation, supporting the historical "women and children first" protocol used during the disaster.

Q3: Survival Rate by Passenger Class



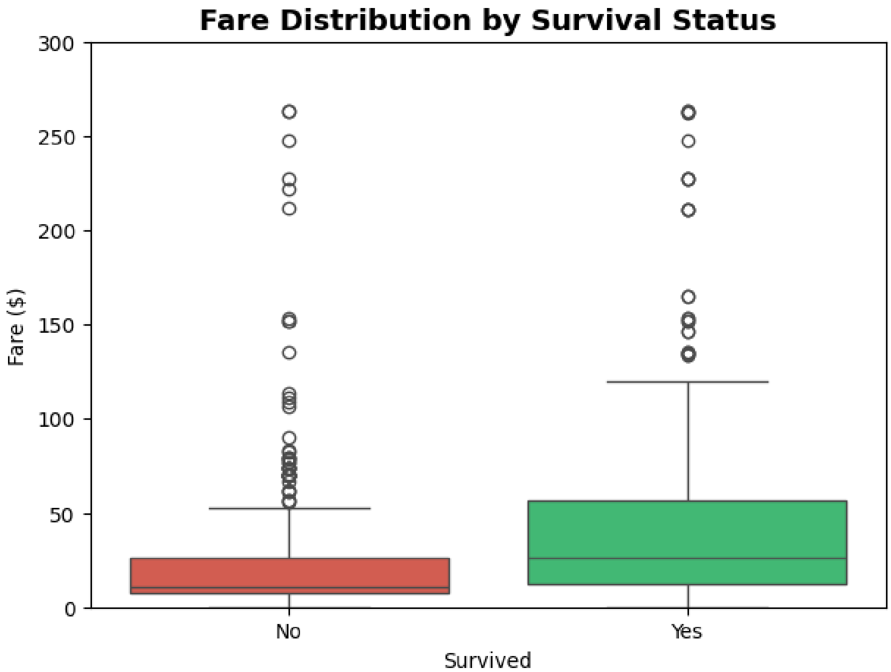
First class passengers survived at the highest rate (63.0%), followed by second class (47.3%); third class had the lowest survival rate (24.2%), pointing to unequal lifeboat access by ticket class.

Q4: Age Distribution by Survival Status



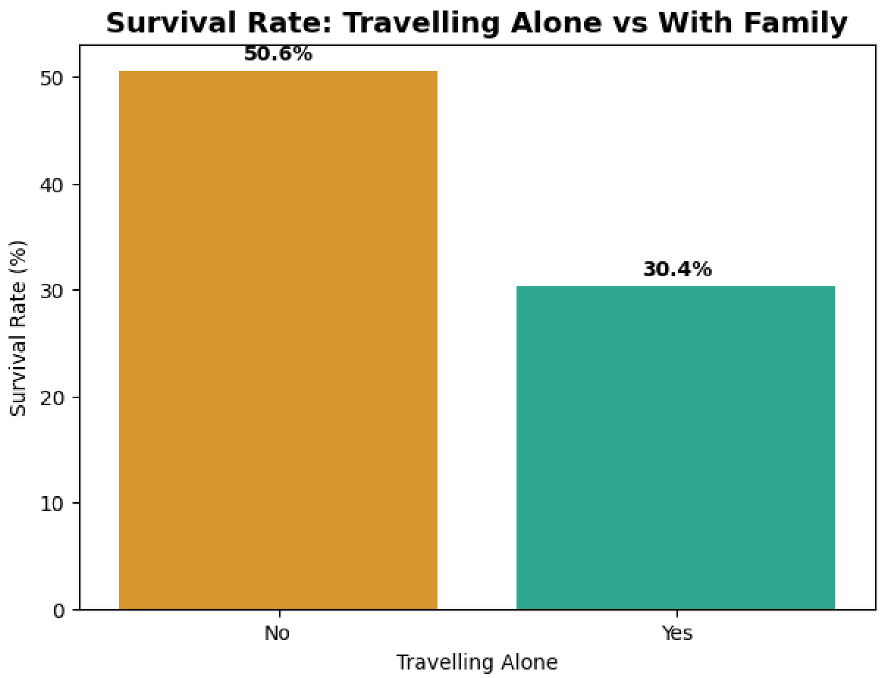
Young children (under 13) had the highest survival rate among age groups, while senior citizens had the lowest — consistent with children being prioritized during evacuation.

Q5: Fare Distribution by Survival Status



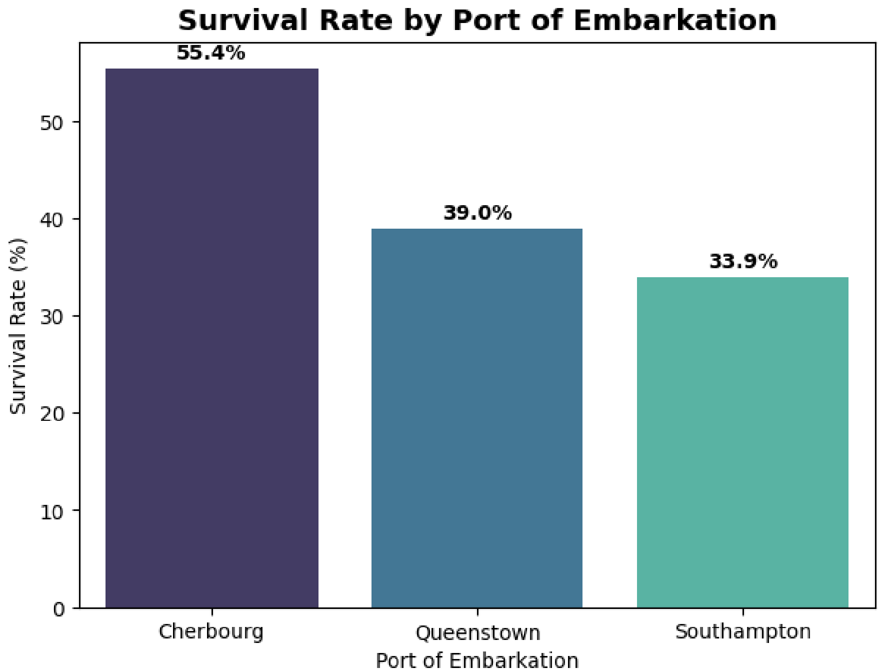
Passengers who survived paid a noticeably higher average fare than those who did not, reinforcing that wealthier passengers had a survival advantage.

Q6: Travelling Alone vs With Family



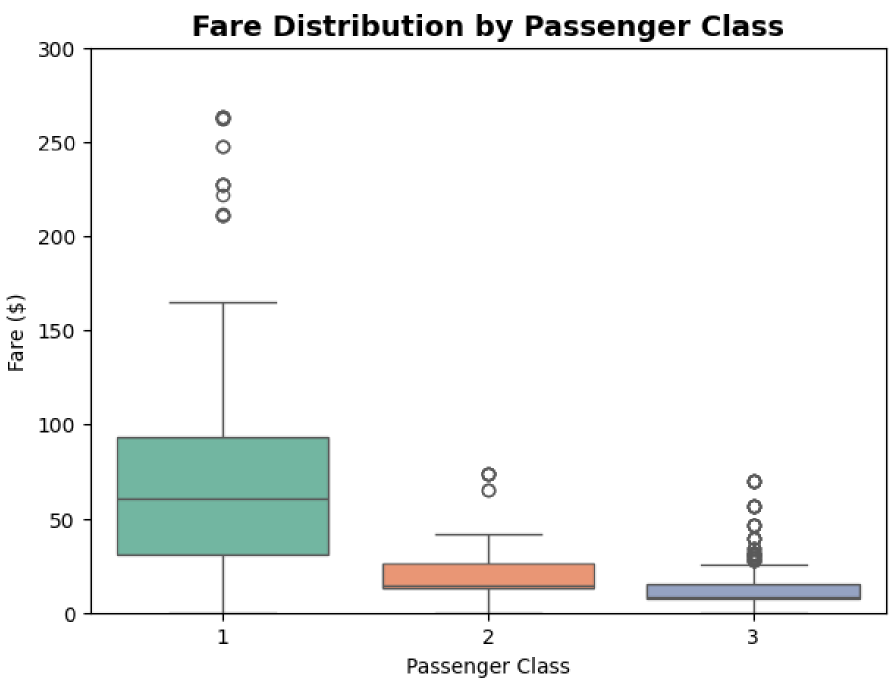
Passengers travelling with family had a higher survival rate (50.6%) than those travelling alone (30.4%). This suggests that travelling with family may have provided support during evacuation, while solo travelers had comparatively lower chances of survival.

Q7: Survival Rate by Port of Embarkation



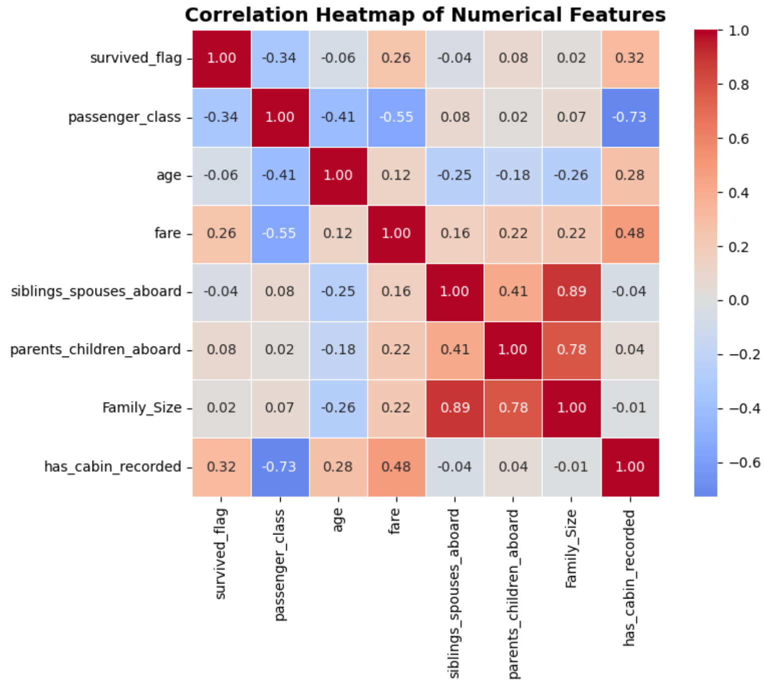
Passengers boarding at Cherbourg (C) had the highest survival rate (55.4%), followed by Queenstown (39.0%), while Southampton (33.9%) had the lowest. This difference may be partly explained by Cherbourg having a higher proportion of first-class passengers, who generally had higher survival rates.

Q8: Passenger Class vs Fare



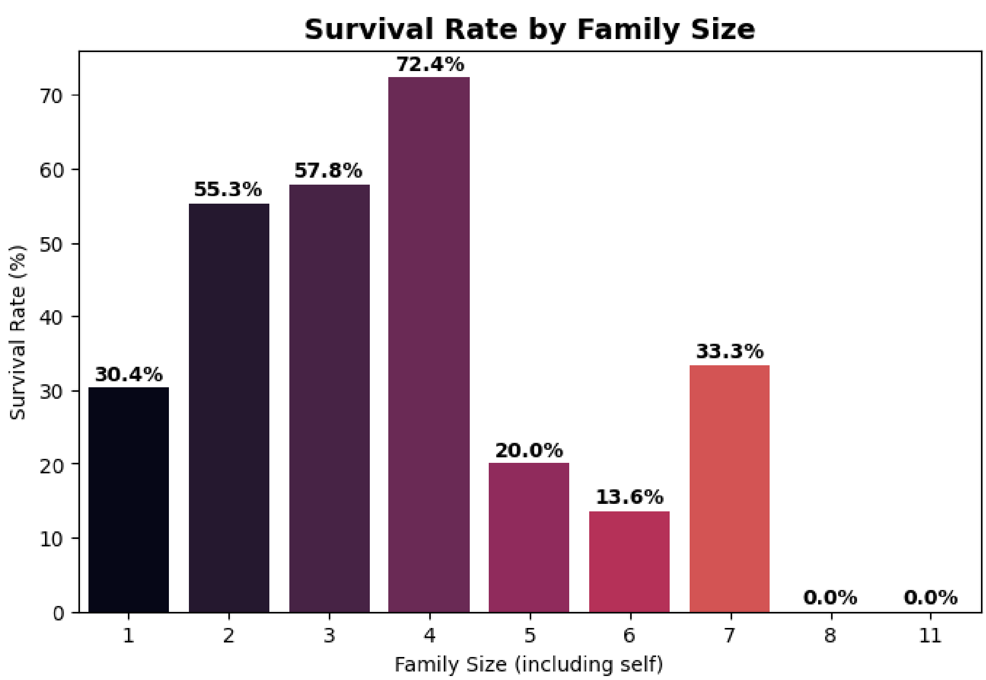
First-class passengers paid significantly higher fares (approximately $84 on average) than second-class passengers (around $21) and third-class passengers (around $14). This confirms a clear relationship between passenger class and ticket price.

Q9: Correlation Heatmap



Strongest positive correlation with survival: has_cabin_recorded (0.32) and fare (0.26). Strongest negative correlation: passenger_class (-0.34), confirming higher-tier tickets (lower class number) meant better survival odds. siblings_spouses_aboard and parents_children_aboard are strongly correlated with each other (0.9) since both feed into family size.

Q10: Additional Analysis — Family Size vs Survival



Survival rate does not increase linearly with family size. Passengers traveling in small groups (family size 2-4) had noticeably higher survival rates than solo travelers, but very large families (size 7+) had sharply lower survival rates — likely because large families struggled to stay together during evacuation.

4. Statistical Analysis

Column	Mean	Median	Mode	Std Dev
Age	29.11	26.00	25.0	13.30
Fare	$32.20	$14.45	$8.05	$49.69
Family Size	1.90	1.00	1	1.61

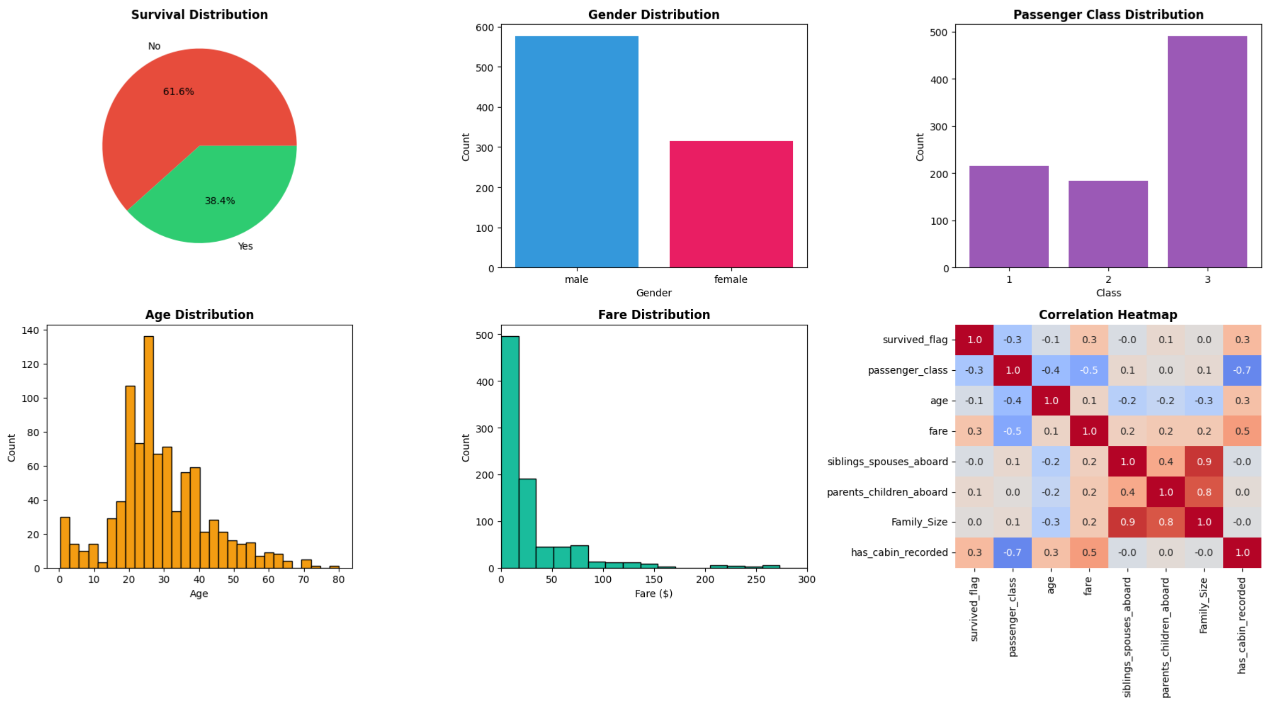
Age: Mean age (~29.1) is higher than the median (26.0), showing a slight right skew — a small number of older passengers pull the average up. The mode (25) shows that young adults were the most common age group aboard. A standard deviation of ~13.3 indicates a fairly wide age spread.

Fare: The mean fare (~$32.2) is much higher than the median (~$14.5), indicating strong right skew — a small number of very expensive first-class tickets inflate the average, while most passengers paid modest fares. The high standard deviation (~$49.7) confirms this large spread, driven by luxury fares.

Family Size: Most passengers traveled with a family size of 1 (i.e., alone) — both the median and mode equal 1. The mean (1.9) being above the median again indicates right skew, caused by a handful of very large families. The standard deviation (~1.6) shows most passengers had small family groups, with a few outliers of large families.

5. Dashboard Summary

A consolidated view of the six most important visualizations from this analysis.



6. Business Insights

1. **Overall Survival Rate:** Only 38.4% of passengers survived, while 61.6% did not, showing how severe the Titanic disaster was.
2. **Gender Differences:** Female passengers had a much higher survival rate (74.2%) than male passengers (18.9%). This suggests that women were given higher priority during the evacuation.
3. **Passenger Class Impact:** First-class passengers had the highest survival rate (63.0%), while third-class passengers had the lowest (24.2%). This shows that passenger class had a strong effect on survival.
4. **Age and Survival:** Children under 13 had the highest survival rate (58.0%), while senior citizens had the lowest (26.9%). This suggests that younger passengers were given more priority during evacuation.
5. **Fare Influence:** Passengers who paid higher fares generally had better survival chances. This is mainly because higher fares were associated with first-class travel.
6. **Family Size:** Passengers travelling with small families generally had better survival rates than those travelling alone or with very large families. Families of 7 or more had noticeably lower survival rates.
7. **Traveling Alone:** Passengers travelling alone had a lower survival rate (30.4%) compared with those travelling with family (50.6%). Family members may have helped each other during the evacuation.
8. **Embarkation Port:** Survival rates differed by embarkation port. Cherbourg had the highest survival rate (55.4%), followed by Queenstown (39.0%) and Southampton (33.9%). This difference may be related to the class distribution of passengers at each port.
9. **Relationship Between Class and Fare:** There was a clear relationship between passenger class and fare. First-class passengers paid the highest fares (around $84 on average), while third-class passengers paid the lowest (around $14), showing that fare was closely related to socio-economic status.
10. **Correlation Analysis:** The correlation analysis showed that no single factor completely determined survival. Survival was influenced by several factors together, including gender, passenger class, age, fare, and family size.

7. Business Recommendations

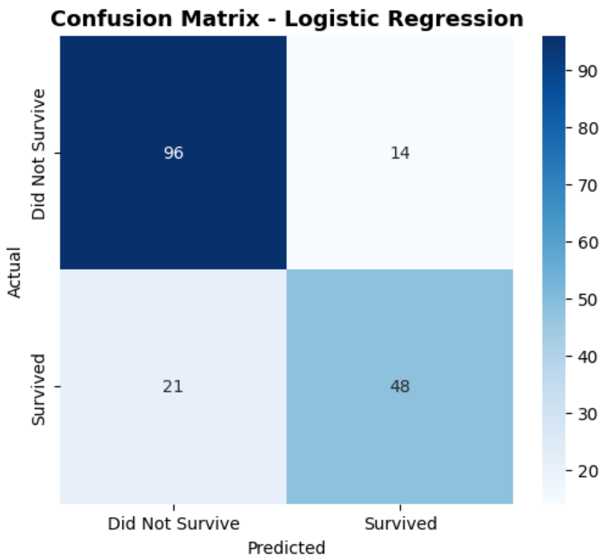
Presenting to the shipping company's management, based on the evidence above:

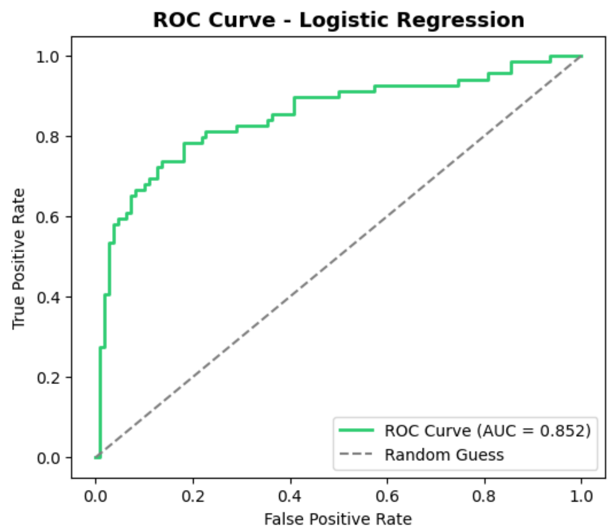
1. **Standardize lifeboat access across all classes.** Since third-class passengers survived at less than half the rate of first-class passengers, ensure lifeboat drills and access routes are equally available regardless of ticket class or cabin deck.
2. **Prioritize family-friendly evacuation protocols.** Since children had the highest survival rate but very large families had the lowest, design evacuation procedures that keep family units together and give trained staff responsibility for large families and elderly passengers.
3. **Improve safety provisions for lower-fare passengers.** The clear fare-survival gap suggests structural issues (cabin location, lifeboat proximity) — invest in evacuation infrastructure serving lower-deck and lower-fare passenger areas.
4. **Provide additional assistance for elderly passengers.** With senior citizens showing the lowest survival rate of any age group, create dedicated mobility assistance and priority evacuation support for older travelers.
5. **Conduct mandatory safety briefings at boarding, tailored by embarkation port and class mix.** Since survival outcomes varied meaningfully by port (likely tied to class distribution), ensure every boarding point delivers the same standard of safety briefing and lifeboat assignment regardless of the typical passenger mix that port serves.

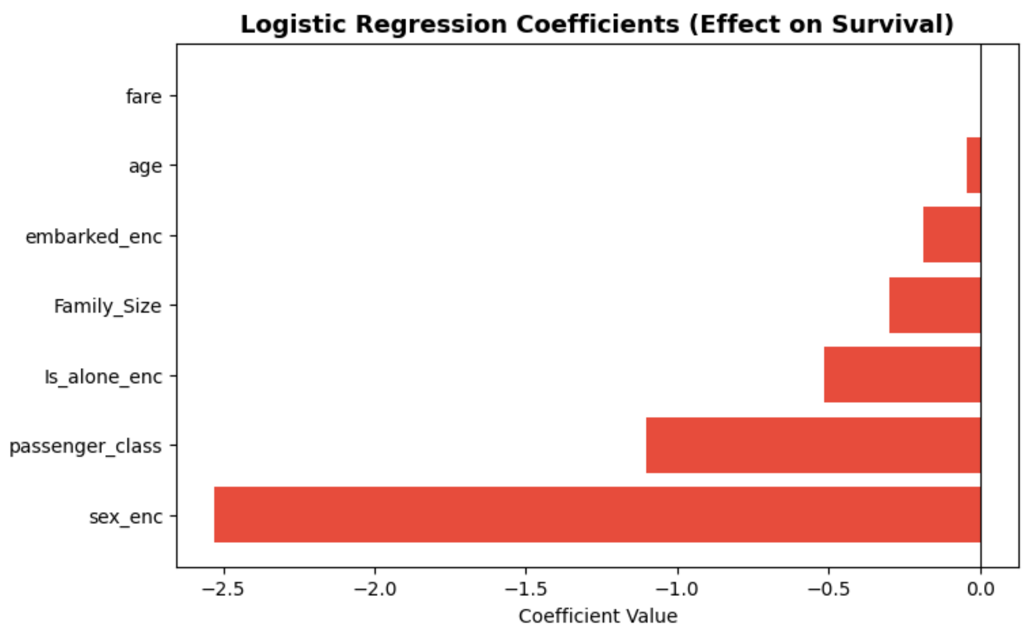
Bonus: Logistic Regression — Predicting Survival

We build a Logistic Regression model to predict whether a passenger survived, using passenger class, sex, age, fare, family size, is-alone status, and port of embarkation as predictors.

Metric	Value
Accuracy	80.4%
AUC Score	0.852







Results & Explanation: The model achieved 80.4% accuracy and an AUC of 0.85 on the held-out test set, meaning it distinguishes survivors from non-survivors quite well using just these 7 features.

Sex has the largest effect by far (coefficient -2.53 for male) — being male sharply reduced the predicted probability of survival, matching the earlier finding that women survived at much higher rates. Passenger class has the second largest effect (-1.10) — moving to a higher class number (lower-tier ticket) reduced survival odds. Is Alone and family size both have negative coefficients, meaning traveling alone or in a larger group reduced predicted survival odds relative to small family groups. Age has a small negative effect, and fare has a small positive effect, both weaker in magnitude once class and sex are accounted for.

This confirms statistically what the exploratory charts suggested visually: gender and class were the dominant drivers of survival, with family structure playing a secondary role.